Slide 1: Executive Overview -- Deterministic Physics First

* Mission: Govern admission, placement, live arbitration, and graceful degradation across 8 coupled physical resource classes (power, cooling, positions, mast, mass_stability, data, compute, time) on a Software-Defined Vessel.

* Core Principle: Hard physical and life/vehicle safety limits are satisfied first; mission capability score, free margin, and displacement cost are optimized second. No priority weight can ever override a hard physical limit or drone ditching deadline.

* Zero Opaque Runtime AI: 100% deterministic, inspectable math (exact cooling curve, integer Watts/kg/bps, parameterized witness explanations, 1 Hz internal simulation stepping).

Slide 2: Fail-Closed Validation & Defeating Legacy Defects

* Strict Schema & Type Discipline (B2, F3, H06, H13, R2, R4):
  * Rejects NaN, Infinity, negative or zero budgets, booleans used as numbers (True != 1), string booleans ("false"), unknown YAML/HTTP JSON fields (UNKNOWN_FIELD), dependency cycles, and invalid cooling coupling (cooling_water_w + cooling_air_mast_w == sustained_power_w).
  * Reproduces and defeats all 10 adversarial input flaws (resource_engine_repro.json) and all 28 verifier mutation classes (false_accepts = 0).
* Six Declarative Guest Cards (G1-G6) with Field-Level Provenance:
  * Official G1 (45 kg, 1.5 kW air), G2 (3,500 kg, 20/6 kW), G3 (5,500 kg, 25/1 kW), G4 (140+850 = 990 kg per errata, 35/10/2 kW), G5 (3,000+25 = 3,025 kg, 45/8 kW), plus custom civil card G6 (420 kg, port_only=true, all fields ASSUMED).

Slide 3: Scenario A (D1) -- 8-Class Admission & Errata Proof

* Organizer Errata Verified (DAT-02, DAT-03):
  * With G4 = 990 kg, exactly 3 of 5 four-guest combinations exceed 10,000 kg: {G1,G2,G3,G4}=10,035 kg, {G1,G2,G3,G5}=12,070 kg, {G2,G3,G4,G5}=13,015 kg.
  * The 2 mass-feasible sets {G1,G2,G4,G5}=7,560 kg and {G1,G3,G4,G5}=9,560 kg prove that passing mass does not imply passing placement: G5’s antenna collides with G1 on mast_3 (1.5+1.2=2.7 kW > 2.0 kW air) and with G4 drone clearance on hangar_roof!
* Single Platform Modification Counterfactual (R6.1):
  * Adding a second guest mast slot mast_4 (<= 60 kg, <= 2.0 kW air) eliminates the G1/G5 mast collision and G4 roof clearance conflict for {G1,G2,G4,G5} and {G1,G3,G4,G5}, leaving only standard Mission Envelope A.10 operational mode scheduling (G5 offline buffering in W4 EMCON and G2/G3/G4 time-multiplexed deck/acoustic windows).

Slide 4: Scenario B (D3) -- Live Arbitration & Causal Acoustic Handshake

* The Conflict: G3 AUV is executing a 40 min quiet acoustic survey (SURVEY_QUIET, 1 kW, 3 kn) when G4 UAV reports 18% battery (< 20% emergency -> 12 min / 720 s max endurance) requiring a turn into 22 kn wind and 35 kW recovery/charging power.
* Causal Command-ACK & Precondition Verification (R1.1, R1.2):
  * D3 approval at T=480 s immediately registers a firm 35 kW reservation (D3_PLAN_V1_G4_RECOVERY) for G4 recovery/charging.
  * Turning into the wind requires a verified ACOUSTIC_HOLD_CONFIRMED ACK from guest_id='G3' matching the active command_id and plan_version, plus power_reserve_w >= 35 kW and deck_clear=True.
  * rescue_primary: Issues ACOUSTIC_HOLD at T=480 s, receives ACK at T=495 s, turns into wind, and lands G4 at T=855 s (345 s margin before T=1,200 s deadline).
  * survey_primary: Completes 240 s of survey pass, issues ACOUSTIC_HOLD at T=720 s, receives ACK at T=735 s, turns into wind, and lands G4 at T=1,095 s (105 s margin before T=1,200 s deadline).

Slide 5: Scenario B (D4) -- Graceful Degradation & Zero Infeasible Ticks

* Honest Power vs. Thermal Accounting (DEG-01..05, R1.1):
  * When generator reserve drops from 200 kW to 100 kW at T=720 s, firm committed power including G3's 25 kW turnaround recharge reservation and G4's 35 kW D3 recovery reservation is 25 + 35 + 45 = 105 kW > 100 kW (5 kW firm deficit).
  * Under both survey_primary and rescue_primary, D4 immediately steps down G5 (45 kW -> 8 kW DEGRADED_70PCT_LOSS) at T=720 s, reducing committed power to 68 kW <= 100 kW (32 kW free margin) and guaranteeing 0 infeasible ticks (91/91 feasible) across the entire mission!
  * At 30 deg C seawater with -40% cooling loop loss (72 kW thermal capacity vs 100 kW power), chilled-water cooling binds first (79.8 kW > 72 kW) and steps down G5 (79.8 kW -> 43.5 kW).

Slide 6: Scenario C -- Crew Regimes (14 / 6 / 0) & Resilience

* Crew Regime & Deck Clearance Enforcement (CRW-01..03, R3, R6.2):
  * 14 Crew: 9 deck crew -> G3 (4 crew) and G4 (3 crew) at-sea vehicle replacement permitted in A' window (4-8 kn, wind <= 20 kn, SS <= 3). Simultaneous G2 winch (0-5 kn) + G4 flight ops (5-12 kn into wind, touching only at 5.0 kn) remain forbidden by Annex A.5 deck_clear and conflicting heading/station-keeping rules.
  * 6 Crew: 2 deck crew -> serialized manual launch/recovery (G2 or G4) permitted with 60-180 s watch handover; at-sea vehicle replacement (min_crew >= 3) blocked.
  * 0 Crew: Zero physical installations or manual deck evolutions at sea (CRW-02); G6 (420 kg > 250 kg, bay install) is port_only=True across all regimes.

Slide 7: Dual-Platform Architecture & Live Ventuno Q Parity

* Decoupled Laptop + Edge Board Execution (B1, DEP-01..03):
  * MacBook Pro (arm64, Python 3.13.15): Full test suite, 1 Hz event-driven simulator, 90 s (20x) compressed demo, and stateful 2D operator console (http://127.0.0.1:8765).
  * Arduino Ventuno Q (training06, Qualcomm Dragonwing IQ8 aarch64, 16 GB RAM, Ubuntu 24.04.4): Live SSH deployment at /home/arduino/sdv-resource-governor.
  * Bit-for-Bit Semantic Parity: Identical campaign_semantic_sha256 verified across arm64 macOS and live aarch64 Ventuno Q (training06).

Slide 8: Single-Screen 2D Operator Console & Fail-Closed Evidence

* Stateful Single-Screen 2D Operator Console (B10, H12, R2, R5):
  * Persistent GovernorSession (session_catalog, manifest_registry, snapshot_id, state_version), per-tick 8-class budget and 2D vessel topology synchronization, 90 s (1000 ms/tick = 20x) Scenario B playback, perturbation injector, and live Scenario A guest card editor.
* Fail-Closed Evidence Verifier (B8, H05, R4):
  * Recomputes all summary metrics and semantic hashes from raw JSON logs, verifies all acceptance criteria <testcase> nodes in tests.* with portable requirements_matrix.csv, cross-checks git_commit and environment, enforces <= 200 ms tick deadlines, and rejects 28 / 28 adversarial mutations (false_accepts = 0).